Notebook 5 — Chapter 5.2

Covariance, Correlation & Conditioning Laws

20 worked examples — covariance algebra, the correlation bound, the Tower Property, and Eve's Law — calibrated for Stage 1/Stage 2 speed and rigor

A. Covariance Algebra & Bilinearity

Example 1 — Covariance From a Joint Table

$X, Y \in \{1,2\}$ with joint PMF $f(1,1)$=0.2, $f(1,2)$=0.1, $f(2,1)$=0.3, $f(2,2)$=0.4. Find $\mathrm{Cov}(X,Y)$.
Solution. $\mathbb{E}[X] = 1(0.3) + 2(0.7) = 1.7$. $\mathbb{E}[Y] = 1(0.5) + 2(0.5) = 1.5$.

$$\mathbb{E}[XY] = 1(0.2) + 2(0.1) + 2(0.3) + 4(0.4) = 2.6$$

$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 2.6 - 1.7(1.5) = \boxed{0.05}$$

⋆ *Analogy:* covariance is a "do they move together?" meter. $\mathbb{E}[XY]$ is what you'd see if X, Y really did move in lockstep; subtracting $\mathbb{E}[X]\mathbb{E}[Y]$ removes the "they just happen to both be positive on average" baseline, leaving only the *genuine* co-movement signal.

Example 2 — Bilinearity Under Affine Transforms

Using Example 1's $\mathrm{Cov}(X,Y) = 0.05$, find $\mathrm{Cov}(3X-2, -2Y+5)$.
Solution. Covariance is bilinear: constants added inside vanish, constants multiplied come straight out:

$$\mathrm{Cov}(aX+b,\, cY+d) = ac \cdot \mathrm{Cov}(X,Y)$$

$$\mathrm{Cov}(3X-2, -2Y+5) = (3)(-2)(0.05) = \boxed{-0.3}$$

▶ *Speed trick:* never re-derive covariance from scratch after a shift/scale — shifts (the -2, the $+5$) are invisible to covariance (it only cares about co-*movement*, not location), and scales just multiply straight through, including their sign.

Example 3 — The Sum/Difference Identity

Using Example 1's $\mathrm{Var}(X) = 0.21$, $\mathrm{Var}(Y) = 0.25$, find $\mathrm{Cov}(X+Y, X-Y)$.
Solution. Expand bilinearly:

$$\mathrm{Cov}(X+Y, X-Y) = \mathrm{Cov}(X,X) - \mathrm{Cov}(X,Y) + \mathrm{Cov}(Y,X) - \mathrm{Cov}(Y,Y)$$

$$= \mathrm{Var}(X) - \mathrm{Var}(Y) = 0.21 - 0.25 = \boxed{-0.04}$$

⋆ *Analogy:* this is the probabilistic twin of the algebraic identity $(a+b)(a-b) = a^2 - b^2$ — the cross terms $\mathrm{Cov}(X,Y)$ and $\mathrm{Cov}(Y,X)$ are identical and cancel, leaving a clean variance difference. A negative result here means Y is the more volatile leg.

Example 4 — Portfolio Variance, Two Assets

Using Example 1's $\mathrm{Var}(X) = 0.21$, $\mathrm{Var}(Y) = 0.25$, $\mathrm{Cov}(X,Y) = 0.05$, find $\mathrm{Var}(X+Y)$.
Solution.

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$$

$$= 0.21 + 0.25 + 2(0.05) = \boxed{0.56}$$

▶ *Analogy:* this is *the* portfolio-risk formula — holding two positively-correlated assets together is *riskier* than the sum of their individual variances would suggest, because the $+2\mathrm{Cov}$ term adds extra joint swing on top. Diversification only helps when that cross term is small or negative (Example 10).

Example 5 — Three-Asset Portfolio, Full Expansion

$\mathrm{Var}(X)$=4, $\mathrm{Var}(Y)$=9, $\mathrm{Var}(Z)$=1. $\mathrm{Cov}(X,Y)$=2, $\mathrm{Cov}(X,Z)$=−1, $\mathrm{Cov}(Y,Z)$=0.5. Find $\mathrm{Var}(X+Y+Z)$.
Solution.

$$\mathrm{Var}(X+Y+Z) = \mathrm{Var}(X) + \mathrm{Var}(Y) + \mathrm{Var}(Z) + 2\big[\mathrm{Cov}(X,Y) + \mathrm{Cov}(X,Z) + \mathrm{Cov}(Y,Z)\big]$$

$$= 4 + 9 + 1 + 2(2 - 1 + 0.5) = 14 + 3 = \boxed{17}$$

⋆ *Analogy:* every pair of assets in the book contributes its own cross term — a 3-asset book has $\binom{3}{2}$=3 pairwise covariances, a 10-asset book has $\binom{10}{2}$=45. This is exactly why real risk desks track a full covariance *matrix*, not just individual variances: the off-diagonal terms are where portfolio risk actually hides.

B. Correlation & the $[-1, 1]$ Bound

Example 6 — Computing ρ From Raw Moments

Using Example 1's numbers ($\mathrm{Cov} = 0.05$, $\mathrm{Var}(X) = 0.21$, $\mathrm{Var}(Y) = 0.25$), find $\rho_{X,Y}$.
Solution.

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{0.05}{\sqrt{0.21}\sqrt{0.25}} = \frac{0.05}{\sqrt{0.0525}} = \boxed{\frac{1}{\sqrt{21}} \approx 0.218}$$

▶ *Analogy:* correlation is covariance with its *units stripped off* — dividing by $\sigma_X \sigma_Y$ rescales the raw co-movement number into a pure, unit-free gauge between -1 and 1, so a \$-denominated covariance and a %-denominated one become directly comparable.

Example 7 — The $\rho = +1$ Boundary Case

$Y = 2X + 3$ for some random X with $\mathrm{Var}(X) > 0$. Show $\rho_{X,Y} = 1$ regardless of X's distribution.
Solution. $\mathrm{Cov}(X, Y) = \mathrm{Cov}(X, 2X+3) = 2\,\mathrm{Var}(X)$ (bilinearity). $\mathrm{Var}(Y) = 4\,\mathrm{Var}(X) \Rightarrow \sigma_Y = 2\sigma_X$.

$$\rho_{X,Y} = \frac{2\,\mathrm{Var}(X)}{\sigma_X \cdot 2\sigma_X} = \frac{2\,\mathrm{Var}(X)}{2\,\mathrm{Var}(X)} = \boxed{1}$$

⋆ *Analogy:* $\rho = 1$ is the signature of a *perfect positive linear* relationship — not "highly related," but "one is exactly an increasing straight-line function of the other." No noise, no curvature allowed; this is the tightest possible co-movement.

Example 8 — The $\rho = -1$ Boundary Case

$Y = -3X + 1$. Find $\rho_{X,Y}$.
Solution. $\mathrm{Cov}(X, Y) = -3\,\mathrm{Var}(X)$; $\mathrm{Var}(Y) = 9\,\mathrm{Var}(X) \Rightarrow \sigma_Y = 3\sigma_X$.

$$\rho_{X,Y} = \frac{-3\,\mathrm{Var}(X)}{\sigma_X \cdot 3\sigma_X} = \boxed{-1}$$

▶ *Analogy:* the negative sign of the slope (-3) passes straight through to ρ, but its *magnitude* (3) cancels out entirely — correlation only cares about direction and tightness of the linear link, never about scale. $Y = -3X + 1$ and $Y = -300X + 1$ both give $\rho = -1$.

Example 9 — Zero Correlation, Continuous Dependence

$X \sim \mathrm{Uniform}(-1, 1)$, $Y = X^2$. Show $\mathrm{Cov}(X,Y) = 0$ yet X, Y are dependent.
Solution. $\mathbb{E}[X] = 0$ by symmetry. $\mathbb{E}[XY] = \mathbb{E}[X^3] = \int_{-1}^{1} x^3 \cdot \frac{1}{2}\,dx = 0$ (odd function over a symmetric interval).

$$\mathrm{Cov}(X,Y) = 0 - 0 \cdot \mathbb{E}[Y] = \boxed{0}, \qquad \rho_{X,Y} = \boxed{0}$$

Yet Y is a *deterministic function* of X — about as dependent as two variables can be.
⋆ *Analogy:* the continuous twin of Ch. 5.1's $Y = X^2$ trap. A symmetric "U-shaped" relationship has exactly as much upward co-movement as downward, so the linear detector (correlation) reads a flat zero while the true relationship — a parabola — is as strong as it gets. **Always plot before you trust a correlation of 0.**

Example 10 — The Minimum-Variance Hedge Ratio

$\mathrm{Cov}(X,Y) = 6$, $\mathrm{Var}(Y) = 4$, $\mathrm{Var}(X) = 20$. Find the hedge ratio h minimizing $\mathrm{Var}(X - hY)$, and the resulting minimized variance.
Solution. $\mathrm{Var}(X - hY) = \mathrm{Var}(X) - 2h\,\mathrm{Cov}(X,Y) + h^2\mathrm{Var}(Y)$. Differentiate in h, set to 0:

$$-2\,\mathrm{Cov}(X,Y) + 2h\,\mathrm{Var}(Y) = 0 \;\Rightarrow\; h^\star = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(Y)} = \frac{6}{4} = \boxed{1.5}$$

$$\mathrm{Var}(X - h^\star Y) = \mathrm{Var}(X) - \frac{\mathrm{Cov}(X,Y)^2}{\mathrm{Var}(Y)} = 20 - \frac{36}{4} = \boxed{11}$$

▶ *Analogy:* this is the textbook options/futures *hedge ratio* — for every 1 unit of Y you short against your X position, you cancel the most correlated risk possible. The minimized variance $20 \to 11$ is the "diversification" promise from Example 4, now run in reverse: a well-chosen

negative-effective exposure strictly reduces risk, and the leftover 11 is the risk correlation alone cannot hedge away.

C. Law of Total Expectation (Tower Property)

Example 11 — The Tower Property, Discrete Regimes

A trade's size X depends on regime $Y \in \{0,1\}$: $\mathbb{P}(Y{=}0){=}0.4$ with $\mathbb{E}[X \mid Y{=}0]{=}10$; $\mathbb{P}(Y{=}1){=}0.6$ with $\mathbb{E}[X \mid Y{=}1]{=}20$. Find $\mathbb{E}[X]$.
Solution.

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid Y]\big] = \sum_y \mathbb{P}(Y{=}y)\,\mathbb{E}[X \mid Y{=}y]$$

$$= 0.4(10) + 0.6(20) = \boxed{16}$$

⋆ *Analogy:* the Tower Property is a *weighted average of averages* — exactly how a school computes an overall pass rate from each class's pass rate, weighted by class size. You never need the full joint distribution of (X, Y), only the regime probabilities and each regime's conditional mean.

Example 12 — Tower Property, Continuous

$Y \sim \text{Uniform}(0,1)$, and $X \mid Y{=}y \sim \text{Uniform}(0,y)$ (Ch. 5.1's triangular structure). Find $\mathbb{E}[X]$.
Solution. $\mathbb{E}[X \mid Y{=}y] = y/2$. Then

$$\mathbb{E}[X] = \mathbb{E}\left[\frac{Y}{2}\right] = \frac{\mathbb{E}[Y]}{2} = \frac{1/2}{2} = \boxed{\frac{1}{4}}$$

▶ *Speed trick:* once $\mathbb{E}[X \mid Y{=}y]$ is a simple function of y, the Tower Property collapses to *taking the expectation of that function* — here $\mathbb{E}[Y/2]$, no new integral over the joint density required, even though the underlying density is the same triangular shape from Ch. 5.1 Example 13.

Example 13 — Random Sums: Wald's Identity

A trader executes a random number of trades N with $\mathbb{E}[N] = 5$; each trade's iid payoff has $\mathbb{E}[X_i] = 2$. Find $\mathbb{E}[S]$ where $S = \sum_{i=1}^{N} X_i$.
Solution. Condition on N: $\mathbb{E}[S \mid N{=}n] = n\,\mathbb{E}[X_i]$ (sum of n iid terms). By the Tower Property:

$$\mathbb{E}[S] = \mathbb{E}\big[\mathbb{E}[S \mid N]\big] = \mathbb{E}[N \cdot \mathbb{E}[X_i]] = \mathbb{E}[N]\,\mathbb{E}[X_i] = 5(2) = \boxed{10}$$

⋆ *Analogy:* a random-length conga line — you don't know how many dancers will join (N is random), but each dancer adds the same expected amount of "length" ($\mathbb{E}[X_i]$) independent of how many came before. Total expected length is just (expected dancers) × (expected length each) — this is **Wald's identity**, the Tower Property's signature application in queueing and trade-flow modeling.

Example 14 — Expected P&L Across Market Regimes

$\mathbb{P}(\text{bull}){=}0.3$, $\mathbb{E}[\text{PnL} \mid \text{bull}]{=}\$50$; $\mathbb{P}(\text{bear}){=}0.2$, $\mathbb{E}[\text{PnL} \mid \text{bear}]{=}-\$30$; $\mathbb{P}(\text{flat}){=}0.5$, $\mathbb{E}[\text{PnL} \mid \text{flat}]{=}\$5$. Find $\mathbb{E}[\text{PnL}]$.
Solution.

$$\mathbb{E}[\text{PnL}] = 0.3(50) + 0.2(-30) + 0.5(5) = 15 - 6 + 2.5 = \boxed{\$11.50}$$

▶ *Analogy:* a desk's year-end expected P&L is never computed by pretending only one regime happens — it's the Tower Property run on market state, exactly Example 11's mechanism with trading-desk labels instead of abstract regimes 0/1.

Example 15 — Using the Tower Property to Find $\mathbb{E}[X^2]$

For Example 12's setup ($Y \sim \text{Unif}(0,1)$, $X \mid Y{=}y \sim \text{Unif}(0,y)$), find $\mathbb{E}[X^2]$ using $\mathbb{E}[X^2 \mid Y{=}y] = \text{Var}(X \mid Y{=}y) + \big(\mathbb{E}[X \mid Y{=}y]\big)^2$.
Solution. For $\text{Uniform}(0,y)$: $\text{Var}(X \mid Y{=}y) = y^2/12$, $\big(\mathbb{E}[X \mid Y{=}y]\big)^2 = y^2/4$.

$$\mathbb{E}[X^2 \mid Y{=}y] = \frac{y^2}{12} + \frac{y^2}{4} = \frac{y^2}{3}$$

$$\mathbb{E}[X^2] = \mathbb{E}\left[\frac{Y^2}{3}\right] = \frac{\mathbb{E}[Y^2]}{3} = \frac{1/3}{3} = \boxed{\frac{1}{9}}$$

⋆ *Analogy:* the Tower Property isn't just for means — *any* function of X can be pulled through the same "condition, then average" machine. This exact computation is the setup for Example 17's Eve's Law verification: a second moment computed via conditioning is half of the variance decomposition story.

D. Law of Total Variance (Eve's Law) & Capstones

Example 16 — Eve's Law, Verified by Brute Force

Regime $Y{=}0$ (w.p. 0.4): $X \in \{8, 12\}$ equally likely. Regime $Y{=}1$ (w.p. 0.6): $X \in \{15, 25\}$ equally likely. Find $\text{Var}(X)$ via Eve's Law, then verify directly.
Solution. Within-regime: $\text{Var}(X \mid Y{=}0){=}4$, $\text{Var}(X \mid Y{=}1){=}25$. Between-regime: $\mathbb{E}[X \mid Y] \in \{10, 20\}$, and from Example 11, $\mathbb{E}[X] = 16$.

$$\mathbb{E}[\text{Var}(X \mid Y)] = 0.4(4) + 0.6(25) = 16.6$$

$$\text{Var}(\mathbb{E}[X \mid Y]) = 0.4(10^2) + 0.6(20^2) - 16^2 = 280 - 256 = 24$$

$$\text{Var}(X) = 16.6 + 24 = \boxed{40.6}$$

Direct check: the full distribution is $X{=}8, 12, 15, 25$ w.p. $0.2, 0.2, 0.3, 0.3$. $\mathbb{E}[X^2] = 296.6 \Rightarrow \text{Var}(X) = 296.6 - 256 = 40.6$ ✓.
▶ *Analogy:* Eve's Law splits total risk into two *additive*, non-overlapping sources — the "noise within each regime" (16.6, how much X wiggles even once you know the regime) and the "uncertainty about which regime you're in" (24, how much the regime *itself* moves the average). Here the between-regime term is actually the *smaller* piece — within-regime noise dominates.

Example 17 — Eve's Law, Continuous Verification

For Example 12/15's setup, find $\text{Var}(X)$ via Eve's Law and confirm it matches $\mathbb{E}[X^2] - (\mathbb{E}[X])^2$ from Example 15.
Solution.

$$\mathbb{E}[\text{Var}(X \mid Y)] = \mathbb{E}\left[\frac{Y^2}{12}\right] = \frac{\mathbb{E}[Y^2]}{12} = \frac{1/3}{12} = \frac{1}{36}$$

$$\text{Var}(\mathbb{E}[X \mid Y]) = \text{Var}\left(\frac{Y}{2}\right) = \frac{\text{Var}(Y)}{4} = \frac{1/12}{4} = \frac{1}{48}$$

$$\text{Var}(X) = \frac{1}{36} + \frac{1}{48} = \frac{4}{144} + \frac{3}{144} = \boxed{\frac{7}{144}}$$

Matches Example 15's $\mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{1}{9} - \frac{1}{16} = \frac{7}{144}$ exactly ✓.
⋆ *Analogy:* two completely different computational routes — (a) raw second moment via the Tower Property, or (b) the two-piece Eve's Law decomposition — land on the *same* number. In a timed exam, whichever route has cleaner intermediate fractions is the one to take; Eve's Law is usually faster once conditional variance has a known closed form (here, Uniform's $M^2/12$).

Example 18 — Compound Variance for a Random Sum

N has $\mathbb{E}[N]{=}5$, $\text{Var}(N){=}3$; iid trade payoffs have $\mathbb{E}[X_i]{=}2$, $\text{Var}(X_i){=}4$, independent of N. Find $\text{Var}(S)$ where $S = \sum_{i=1}^{N} X_i$.
Solution. Condition on N: $\mathbb{E}[S \mid N] = N\mathbb{E}[X_i]$, $\text{Var}(S \mid N) = N\,\text{Var}(X_i)$ (sum of N independent terms). By Eve's Law:

$$\text{Var}(S) = \underbrace{\mathbb{E}[N\,\text{Var}(X_i)]}_{\mathbb{E}[N]\text{Var}(X_i)} + \underbrace{\text{Var}(N\,\mathbb{E}[X_i])}_{\text{Var}(N)\,\mathbb{E}[X_i]^2}$$

$$= \mathbb{E}[N]\text{Var}(X_i) + \text{Var}(N)\big(\mathbb{E}[X_i]\big)^2 = 5(4) + 3(4) = 20 + 12 = \boxed{32}$$

▶ *Analogy:* total P&L variance from a random-length trading day has *two* risk sources stacked exactly like Example 16 — the per-trade noise, scaled by how many trades typically happen ($\mathbb{E}[N]\text{Var}(X_i)$), plus the extra risk purely from *not knowing the trade count itself* ($\text{Var}(N)\mathbb{E}[X_i]^2$). Even with zero per-trade risk, an uncertain N alone creates P&L variance.

Example 19 — Within-Regime Risk vs. Between-Regime Risk

Using Example 14's regimes, with $\text{Var}(\text{PnL} \mid \text{bull}){=}100$, $\text{Var}(\text{PnL} \mid \text{bear}){=}225$, $\text{Var}(\text{PnL} \mid \text{flat}){=}16$. Find $\text{Var}(\text{PnL})$ and compare the two risk sources.
Solution.

$$\mathbb{E}[\text{Var}(\text{PnL} \mid \text{regime})] = 0.3(100) + 0.2(225) + 0.5(16) = 83$$

$$\text{Var}(\mathbb{E}[\text{PnL} \mid \text{regime}]) = 0.3(50^2) + 0.2(-30^2) + 0.5(5^2) - 11.5^2$$

$$= 942.5 - 132.25 = 810.25$$

$$\text{Var}(\text{PnL}) = 83 + 810.25 = \boxed{893.25}$$

⋆ *Full-circle intuition:* **not knowing which regime you're in** (810.25) swamps the P&L noise *within* any single regime (83) by nearly 10× — the opposite balance from Example 16. This is exactly why systematic desks spend more effort on regime *detection* than on reducing trade-level noise: the dominant risk source is uncertainty about the macro state itself.

Example 20 — Full Capstone: Tower, Eve's Law & the Law of Total Covariance

Hidden regime $Y \in \{0,1\}$, $\mathbb{P}(Y{=}0) = \mathbb{P}(Y{=}1) = 0.5$. Given $Y{=}0$: $(X,Z) \in \{(8,4),(12,6)\}$ equally likely. Given $Y{=}1$: $(X,Z) \in \{(18,15.5),(22,14.5)\}$ equally likely. Find (a) $\mathbb{E}[X]$, (b) $\mathrm{Var}(X)$, (c) $\mathrm{Cov}(X,Z)$.

Solution. (a) *Tower:* $\mathbb{E}[X \mid Y{=}0]{=}10$, $\mathbb{E}[X \mid Y{=}1]{=}20 \Rightarrow \mathbb{E}[X] = 0.5(10) + 0.5(20) = \boxed{15}$. (b) *Eve's Law:* $\mathrm{Var}(X \mid Y{=}0) = \mathrm{Var}(X \mid Y{=}1) = 4 \Rightarrow \mathbb{E}[\mathrm{Var}(X \mid Y)] = 4$. $\mathrm{Var}(\mathbb{E}[X \mid Y]) = 0.5(10^2){+}0.5(20^2) - 15^2 = 250 - 225 = 25$. So $\mathrm{Var}(X) = 4 + 25 = \boxed{29}$. (c) *Law of Total Covariance* (same split, applied to a pair):

$$\mathrm{Cov}(X,Z) = \mathbb{E}\big[\mathrm{Cov}(X,Z \mid Y)\big] + \mathrm{Cov}\big(\mathbb{E}[X \mid Y], \mathbb{E}[Z \mid Y]\big)$$

$\mathrm{Cov}(X,Z \mid Y{=}0){=}2$, $\mathrm{Cov}(X,Z \mid Y{=}1){=}{-}1 \Rightarrow \mathbb{E}[\cdot] = 0.5(2){+}0.5(-1) = 0.5$. $\mathbb{E}[Z \mid Y{=}0]{=}5$, $\mathbb{E}[Z \mid Y{=}1]{=}15$, $\mathbb{E}[Z] = 10$: $\mathrm{Cov}(\mathbb{E}[X \mid Y], \mathbb{E}[Z \mid Y]) = \big[0.5(10)(5){+}0.5(20)(15)\big] - 15(10) = 175 - 150 = 25$.

$$\mathrm{Cov}(X,Z) = 0.5 + 25 = \boxed{25.5}$$

(Direct brute-force computation on the full 4-point joint confirms all three results exactly.)

▶ *Full-circle intuition:* the entire chapter, run back-to-back on one hidden-regime model — the Tower Property for the mean, Eve's Law for the variance, and its natural extension (the Law of Total Covariance) for the covariance between *two* regime-linked quantities. In all three, risk/co-movement splits into a "within-regime" piece and a "because-the-regime-itself-varies" piece — and in every case here, **the hidden regime is the dominant source**, exactly the lesson a real trading desk learns about latent market states.